Real case - Exercice

MesoNH Tutorial - 2-5 November 2026

Presentation

Objectives :

- run a MESONH simulation in real case
- understand the different steps with 1 or 2 models
- discover and modify the namelists

Please :

Modify only what is asked in the namelists

Preparation

cd ~/MNH-V6-0-1/examples
mkdir TP_CAS_REEL
cd TP_CAS_REEL
tar xvf ~rodierq/tp_real_makefile.tar
export PREP_PGD_FILES=~rodierq/PREP_PGD_FILES_WWW

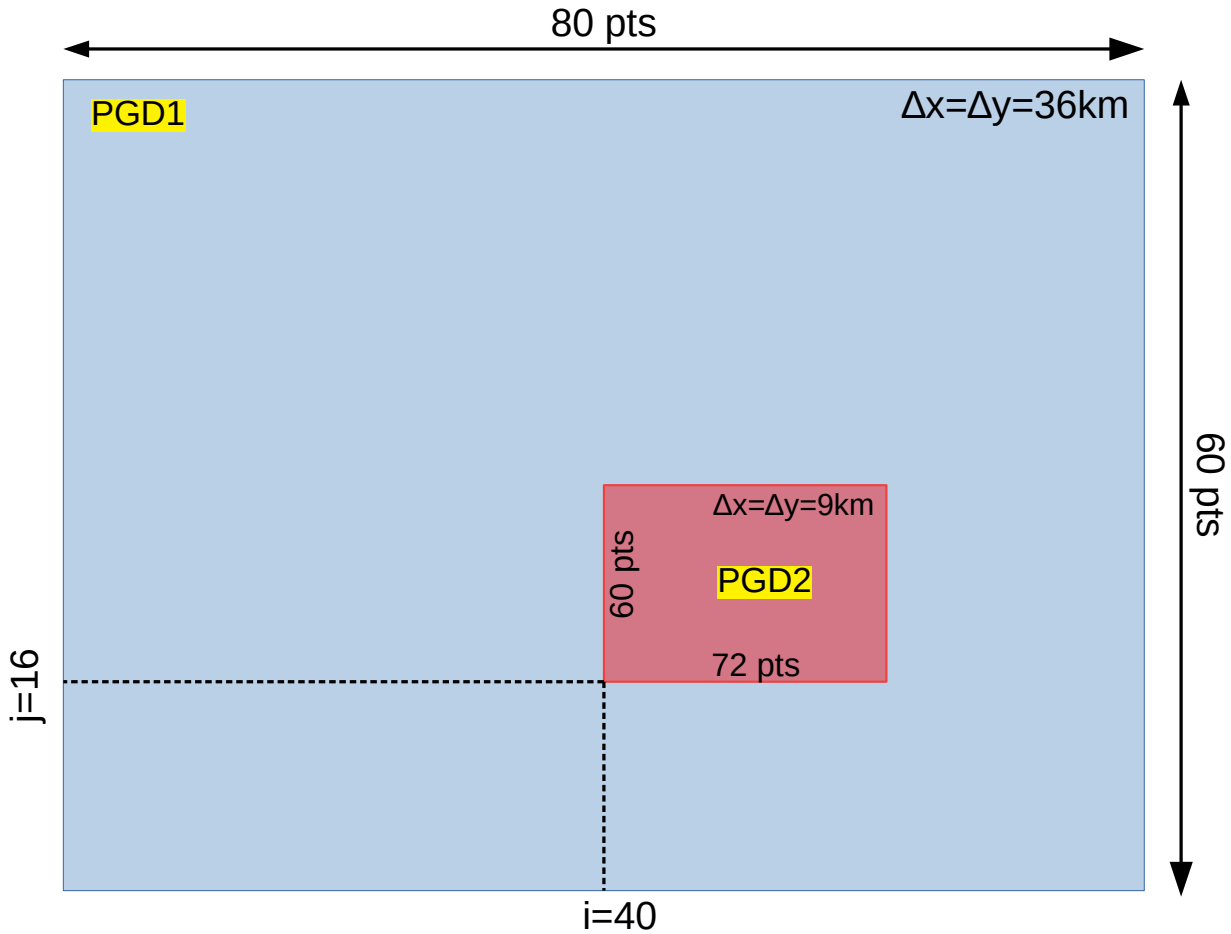
For each new working terminal

Load the profile :
source ~/MNH-V6-0-1/conf/profile_mesonh-LXgfortran-R8I4-MNH-V6-0-1-MPIAUTO-DEBUG

Creation of all the PDG files

1. In the directory `001_pgd1`, run the step PREP_PGD to create the **parent**'s PGD file named **PGD_36km** with :
   - a domain with 80 points in x-direction and 60 in y-direction
   - a mesh of 36 km in x and y
2. In the directory `002_pgd2`, run the step PREP_PGD to create the **child**'s PGD file named **PGD_9km** with :
   - a domain with 72 points in x-direction and 60 in y-direction (number of points for the child's domain)
   - a mesh of 9 km in x and y
   - which start at point i=40,j=16 from parent's domain
3. In the directory `003_nest`, run the step PREP_NEST_PGD

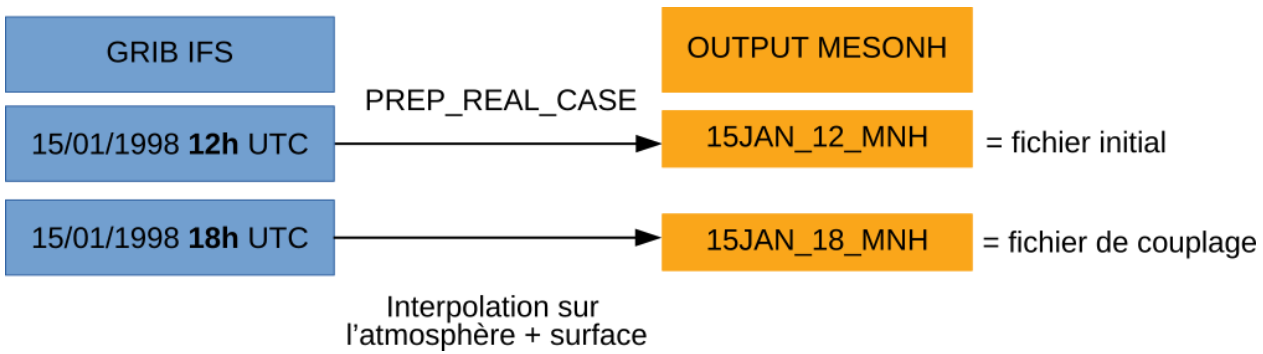
Creation of all the PGD files



Preparation of coupling files

The extraction of the GRIB files was already done. The files are in the directory 004_prep_real

1. in the directory 004_prep_real run the step PREP_REAL_CASE to make the inital file for the parent's domain with the data file **ecmwf.EI.19980115.12**. Name the output as **15JAN_12_MNH**
2. run the step PREP_REAL_CASE to make the coupling file for the parent's domain from the data file **ecmwf.EI.19980115.18**. Name it **15JAN_18_MNH**



Segment 1

We will now run the simulation with only one domain between 12h and 13h with a coupling file at 18h.

1. In the directory `005_run1`, modify the namelist to have :
   - 1 domain
   - 1 hour of simulation
   - 4 backup files (every 15 minutes)
   - a time step of 100 s
   - the backup files must be named : **16J36.1.SEG01.00n**
2. Run the MESONH simulation

Segment 2

1. In the directory `005_run1`, modify the namelist in order to restart the simulation for 1 hour. The backup files must be named : **16J36.1.SEG02.00n**
2. Run the MESONH simulation

Segment 3

Now, run a simulation with the 2 domains between 14h and 15h. First, create the initial file for the child's domain.

1. In the directory `006_spa_mod1_mod2`, run the step SPAWNING (modify the namelist) to make the horizontal interpolation from the parent's domain to the child's domain at 14h (end of segment 2)
2. In the directory `007_preal`, modify the namelist in order to create the child's initial file named **15JAN_14_MNH2** (vertical interpolation after SPAWNING)

Segment 3

3. In the directory `008_run2`, modify the namelist to have :
   - 2 domains
   - 1 hour of simulation
   - 2 backup files (every 30 minutes) for each domain
   - a time step of 100 seconds for the parent and a ratio of 4 with its child
   - two-way nesting interaction
   - the backup files must be named : **16J36.1.SEG03.00n**
4. Run the MESONH simulation
5. In the directory `009_diag`, run the step DIAG on the files you want and plot some figures in `010_python`

Bonus

If you have finished early, here are some harder exercices.

3 domains

- add a 3rd domain located at the top-left of PGD1 with a 18km resolution (choose the domain size)
- run a simulation with the 3 domains for 2 hours starting at 12 UTC the 15/01/1998 with a frequency output of 1 hour.

Run domain 3 alone

- run a simulation with the domain 3 alone (coupled with domain 1 without grid-nesting) starting from the result of the previous simulation at **13** UTC the 15/01/1998 for one hour long